

Grounded Symbols: $\{task_objects, target_locations\}$

Domain.pddl

Deliberation & Plan Verification (LLaMA 3 + Fast Downward)

PDDL Synthesizer

Problem Formulation
(Structured JSON)

Auto-Correction

Error Feedback Loop
(up to n retries)

Domain Validator

Pydantic Schema &
Regex Consistency Check

Fast Downward

Classical PDDL Planner

Action Plan

$\pi = \langle \text{pick, place, } \dots \rangle$

JSON Problem

Invalid

Valid: *problem.pddl*

-Reprompt-